Performance Gap Analysis of Global Renewable Energy (SDG 7 - 7.2 clean global energy)

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Abstract

The transition towards sustainable and clean global energy has become one of the major priority for the United Nations and their Sustainable Goal particularly focusing on the SDG 7 (clean global energy) and SDG 7.2 [1] which aims to increase the global share of renewable energy production infrastructure and services such as solar, wind, hydro-power instead of relying solely on fossil fuels, these alternative energy source are designed to be clean, cheaper naturally obtained and closer to zero emission . Even though the renewable resource related systems and infrastructure to obtain clean energy are growing significantly but there is a critical gap exists between installed capacity and the actual generation where the installed capacity focuses on the overall power delivery of a particular clean-energy generating system , while actual capacity is the actual power that has been transferred energy using the clean-energy generating systems. Our study investigates the discrepancy using data analytics driven approach focusing on global time-series related data starting from 2000 to 2023 , Using our SOTA (state of the art) data analytics and statistical approaches we are able to identify key inefficiencies in the production of renewable energy and its overall utilization , including grid congestion , temporal mismatches between supply and demand insufficient infrastructure development and improper power generation based on regions and time stamps and analyzing overall consumption of fossil fuels comparing it with the use of clean energy utilization Using statistical analysis , time series modeling classification , predictive techniques the study analyzes effectiveness and the performance of renewable energy deployment across different regions and technologies currently relying on. The "performance gap" is introduced to understand and to quantify the divergence between the theoretical energy generation potential and real world output. this will prove the increased infrastructure capacity and power generation capability does not guarantee a reduction in fossil fuel consumption and dependence, emphasizing the importance of grid efficiency and energy storage systems. The study concludes by proposing a data-driven framework for optimization of renewable energy systems and resources, enabling policymakers and stakeholder to make better informed decisions that align with SDG 7.2 targets . This research highlights the critical role of analytics system in transforming renewable energy investments into measurable environmental outcomes

Keywords: SDG 7.2, Clean-energy, fossil-fuel consumption, installed capacity , actual capacity

1 Introduction

The increasing urgency of climate change, environmental degradation excess use of fossil fuels and the depletion of non-renewable resources has made a significant global shift towards investing in sustainable energy system and infrastructure. In response the UN has established Sustainable Development Goal 7 (SDG 7) which mainly focuses on ensuring the global availability of clean, affordable, reliable, sustainable, close to zero-emission energy and modern energy for all by the year 2030 which is proposed in this framework, Target SDG 7.2 [1] specifically emphasizes the overall expansions of renewable energy sources in the global energy mix. Past two or so decade has shown significant progress in the infrastructure development and deployment of the renewable systems and technologies have had some success on the installation of the infrastructure mainly solar and wind. Focusing on the growth has not been a favourable part mainly because of the lack of consistency and lack of renewable infrastructure utilization and maintenance thus falling back to using fossil fuels as the primary source thus raising concerns about the efficiency and effectiveness of current energy transition strategies

The major challenges lies between the disconnect of installed renewable energy capacity and actual electricity generation, while many countries report on the exponential growth of renewable ecosystem infrastructure alternative energy sources and fuels, still their dependence on fossil fuels still remains unchanged, This phenomenon suggests that the issue is not merely about increasing capacity, but rather about optimizing how that capacity is utilized within the existing energy systems ecosystems and intergrations Factors such as grid limitations, variability inadequate storage solutions in renewable clean energy production has contribute to this inefficiency

Traditional approaches to energy planning often focuses on infrastructure expansion and scalability not without considering proper system design, energy production, energy usage and system integrations. This results in scenerios where renewable energy is generated but cannot be effectively stored, transmitted, or served, leading to energy losses and reduced economic returns. There is a growing need to shift from capacity driven stratgies to data driven decision making frameworks to emphasize the efficiency of the overall system performance

This study address the challenges by adopting a comprehensive data analytics approach to evalute analyze renewable energy systems and performance metrics on a global scale , thus by analyzing data and identifying patterns in energy production and consumption the research aims to uncover the underlying causes of the overall inefficiency in renewable energy systems. The insights generated from this analysis are intended to support more effective energy planning and policy development, ultimately contributing to the successful achievement of SDG 7.2 global scale by evaluting data and processing the patterns in energy production and consumption Our research aims to discover the underlying construct of the overall seriousness of inefficeny in renewable resource energy system and The insights can be generated using data analytics technique which enables you to

2 Literature Review

The global conversation around renewable energy has evolved considerably over the past two decades. What once centered on whether clean energy technologies could work at scale has shifted toward a more nuanced and arguably harder question whether the systems built around those technologies are actually working well. Scholars across engineering, economics, and public policy have increasingly converged on a shared concern, that installing renewable capacity is not the same as delivering renewable energy, and that the gap between the two deserves far more attention than it currently receives [2].

Grid infrastructure has emerged as one of the most persistently cited obstacles in this transition. International energy organizations have repeatedly flagged that transmission networks in many parts of the world were simply not designed with large scale variable renewables in mind [3]. Regions that have moved fastest in deploying solar and wind capacity parts of Europe, India, and the western United States have also experienced some of the highest rates of curtailment, where generated electricity is deliberately wasted because the grid cannot absorb it [4]. This is not a minor technical footnote it represents a real and measurable drag on the returns from renewable investment.

The concept of the capacity factor has become central to understanding this performance gap. Unlike installed capacity, which simply measures how much power a system could theoretically produce, the capacity factor captures how much it actually produces relative to that ceiling [5].A wind farm rated at 500 megawatts that runs at a 30percentage capacity factor delivers the same annual energy as a 150 megawatt plant running continuously and the difference matters enormously when governments and investors are making decisions based on nameplate figures. Researchers have argued that policy frameworks relying primarily on installed capacity statistics risk painting a systematically optimistic picture of energy transition progress [6].

Perhaps no concept better illustrates the operational complexity of renewable integration than the so-called "duck curve," a phenomenon first documented in California and since observed in solar-heavy grids around the world [7]. The shape of the curve named for its resemblance to a duck in profile describes the mismatch between midday peaks in solar generation and the evening surge in consumer demand. As solar penetration deepens, this mismatch sharpens, creating steep ramp requirements for backup generation in the late afternoon hours. The implications reach beyond inconvenience: without storage or demand-response infrastructure, grid operators are forced to rely on fast-ramping fossil fuel plants precisely at the moments when demand is highest, undermining the displacement effect that solar is supposed to achieve [8]. This dynamic also has economic consequences, as it drives investment toward peaker plants that may only operate for a few hours per day, adding costs without advancing decarbonization.

3 Background

3.1 Descriptive Analytics

The Descriptive analystics actually the main foundational tier of the our proposed multi layered analyitics pipleine , our role includes more than summarisation rather it constructs a statistically accurate and temporaly consistent representation of our energy systems has evolved our time based on certain obeservatable parameters and time window , Establishing our empirical baseline is the main prerequiste before reasoning , future projections , resource optimisation can be pursued meaninfully at a minimum thus the basline characterise must focus on the trajectory of the electricity generation , the accumlation of the installed capacity and the rate at which the renewable techolohy and infrastructure has been penerated to the generation mix and the recurring seasonal and structural regularities integrated into the data

3.2 Growth Rate Quantification

The temporal indicators are applicable to the energy systems , The relative growth accumalte over time is a very important analytically indispensable , The growth rate mainly focuses on the change as a fraction of the prior obeseravtion , This is normalisation yields a dimensionless quantity that is directly cross comparable across countries , technology , categories , time horizons , The property is actually very important when the framerowk is actually deployed on the heterogeneous multi country datasets

Let F_t denote any energy related variable of interest such as total electricity generation, installed renewable capacity, or grid transmission volume observed at discrete time index t, where $t = 1, 2, \ldots, T$. The period-on-period relative growth rate is formally expressed as:

$$\text{Growth}_t = \frac{F_t - F_{t-1}}{F_{t-1}}, \quad t = 2, 3, \ldots, T. \tag{1}$$

In Eq. (1), the numerator captures absolute increment between consecutive periods, while the denominator normalises against the magnitude of the base period. A positive value signals expansion in the system variable; a negative value indicates contraction Applying Eq. (1) across the full observation window produces a growth profile that faithfully traces the dynamic trajectory of the energy system over time.

3.3 Renewable Energy Share

The growth rate characterises influence the change in each variables , it doesnt speak or implies directly to the structural composition of the energy mix analysing the extend to which a system has recovered and moved away from fossil fuel dependence requires a special separate indicator : The renewable energy share will be computer as a fractional measures of the systems green generation intensity at each point in time

This metric also be defined as the ratio of electricity produced from renewable sources to the total system generations

$$RS_t = \frac{G_t^{\mathrm{ren}}}{G_t^{\mathrm{tot}}}, \tag{2}$$

The G_t^{ren} denotes electricity produced from renewable technologies like for example solar,wind,hydroelectricty at time t, and G_t^{tot} denotes total generation from all sources including fossil fuels. The resulting metric $RS_t \in [0, 1]$ provides a bounded, scale-invariant measure robust to differences in the absolute size of the systems under comparison. A value approaching unity indicates near complete renewable dominance, while values below 0.20 are conventionally associated with high fossil fuel dependence rathe independence from fossil fuel. The time series $\{RS_t\}_{t=1}^{T}$ serves as one of the primary monitoring indicators for tracking energy transition progress.

3.4 Additive Time-Series Decomposition

Moving the average smoothing techiques actually isolates the trend signal structure but leaves its constituent structural forces can be entangkes , In real time energy system the obeserved generation and consumption patterns can be simultaneously shaped by three distinct systems , a long run secular trend driven by capacity expansions and technological change , a periodic seasonal componet that reflect recurring cycle of climatological and behaviour , and an irregular residual componets that actually captring the idiosyncratic shocks , measurment error and unexplainable phenomenons . Disentangling these forces ia analytically essential, since conflating them leads to misattribution of casual drivers and distored views

The additive decomposition model formalises this separation by expressing each observation Y_t as the sum of its component parts:

$$Y_t = T_t + S_t + \varepsilon_t, \tag{3}$$

4 Diagnostic Analytics: Causal and Structural Inference

4.1 Overview

The descriptive analytics establishes , what has occurred in particular energy system and its infrastructure , but the diagnostic analytics takes on more elaborative task of explaining why it has occurred , This layer requires intensive information beyond just basic summary statistical and trend analysis to examine we need go beyond classical statistical approach move into analyzing the structural mechanisms, relationships

Algorithm 1 Descriptive Analytics Pipeline

Require: Time series $\{F_t\}_{t=1}^{T}$, generation arrays $\{G_t^{\text{ren}}\}$, $\{G_t^{\text{tot}}\}$, smoothing window n, seasonal period s

Ensure: Growth rates $\{\text{Growth}_t\}$, renewable share $\{RS_t\}$, smoothed series $\{MA_t\}$, decomposition triplets $\{(T_t, S_t, \varepsilon_t)\}$

for $t = 2$ **to** T **do**
  $\text{Growth}_t \leftarrow (F_t - F_{t-1})/F_{t-1}$ ▷ Eq. (1)
  $RS_t \leftarrow G_t^{\text{ren}}/G_t^{\text{tot}}$ ▷ Eq. (2)
end for
for $t = n$ **to** T **do**
  $MA_t \leftarrow \frac{1}{n}\sum_{i=t-n+1}^{t} F_i$ ▷ Eq. (**??**)
end for
Estimate $\{T_t\}$ via LOESS regression on $\{(t, Y_t)\}$
Compute detrended series: $D_t \leftarrow Y_t - T_t$ for all t
for each seasonal phase $p = 1$ **to** s **do**
  $S_p \leftarrow \frac{1}{|\mathcal{I}_p|}\sum_{t\in\mathcal{I}_p} D_t$, where $\mathcal{I}_p = \{t : t \equiv p \pmod{s}\}$
end for
Compute residuals: $\varepsilon_t \leftarrow Y_t - T_t - S_t$
Inspect residual ACF; flag if significant autocorrelation remains
return $\{\text{Growth}_t\}$, $\{RS_t\}$, $\{MA_t\}$, $\{T_t, S_t, \varepsilon_t\}$

and latent inefficienceies that underlies observed system behavior , energy system will require simultaneoulsy constrained by the engineering componets physics , infrastructure details nuances , regulatory framework , market dynamics and geophysical componets because of these complex interchangeable variables external factors and componets we cannot rely soley on surface level patterns rarely yield to an straightforward interpretation , The diagnostic layer furnishes the methodlogical apparatus needed to decompose these patterns into their contributing drivers

The diagnostic phase is organised around four complementary analytical instruments. The capacity factor serves as a normalised measure of infrastructure utilisation efficiency, making the explicit the gap between potential and realised output. Pearson correlation analysis maps statistical dependencies between key system variables, enabling detection of substitution effects, complementarities, and structural decouplings. Lagged regression models account for the temporal delays that pervade the energy sector, where the consequences of policy interventions and capital investments typically materialise only after a significant lead time. Finally, K-means clustering partitions heterogeneous multi country or multi regional datasets into internally coherent performance groups, supporting the identification of system archetypes and the targeted diagnosis of bottlenecks within each group. Collectively, these instruments transform the empirical baseline established in Sect. **??** into a body of causal and structural knowledge that informs the predictive modelling developed in Sect. 5.

4.2 Capacity Factor Analysis

The capacity factor is among the most widely employed performance metrics in energy systems engineering. It assesses the degree to which installed generating infrastructure is productively utilised. An energy system may possess substantial nameplate capacity—measured in megawatts (MW)—yet deliver comparatively modest actual generation—measured in megawatt-hours (MWh)—owing to operational constraints such as grid congestion, inadequate transmission infrastructure, curtailment policies, or demand shortfalls. The capacity factor renders this utilisation gap analytically explicit by expressing actual generation as a fraction of the theoretical maximum output that installed capacity would deliver under continuous full-load operation throughout the year.

The capacity factor is among the most popular and widely recognised when it comes to measuring performance metrics and energy system and its architectural performance and overall componets power delivery capacity measurement , It actually assesses the overall dgeree to which the installed generating infraststructure is productively been utilized , An energy system or an infrastructure is capable of handling substantial capacity which has been measures in megawtts (MW , yet deliver only a comparatively modest actual generation measured in megawatts hours (MWh) , due operational constraints external factors such as grid congestion inadequate transmission infrastructure , curtailment policies , in efficient network and maintained infrastructure can contribute to the demand shortfalls or failure in meeting the required power supply transmitting to the grids and other resources . The capacity factor renders this utlisation gap analytically explicit by expressing the actual generation as a fraction of the maximum theoretical output that installed capacity could be delivered under full load continuous through out the year

This is more formally represented as the annual capacity factor

$$CF_t\ (\%) = \frac{G_t^{\text{act}}}{C_t \times 8{,}760} \times 100, \tag{4}$$

where G_t^{act} is actual electricity generation in MWh during year t, C_t is installed capacity in MW at the start of year t, and 8,760 represents the number of hours in a standard year. Values that fall substantially below technology specific benchmarks signal the presence of systemic inefficiencies warranting targeted diagnostic attention.

4.3 Pearson Correlation Analysis

Once we are able to identify the quantified individual system variables and their utilisation efficiency , The next major diagnostic task involves characterising and assesing the statisitical relationship among the paris of variables and their relationship between them what factors they are co related . In energy system and infrastructure , understanding whether the growth in the renewable capacity correlates or coincide with reduction in fossil fuels generations where the economic expansion and impact drives proportional demand increases or whether the grid investment predicts whether a certain technology is feasbile is comparing the corelation analysis of the existing technnology they are using which may be comprised of non renewable resources how easy is to make the switch to a more sustainable resources , whether these system will increase demands , efficiency or whether grid investment predicts gains in operational efficiency is essential for constructing a coherent causal narrative and for avoiding spurious policy prescriptions and grounded in coincidental movements

The Pearson product moment correlation coefficient provides a standardised measure of linear association between two variables. For energy system variables X_t and Y_t observed across T periods, it is defined as:

$$r_{XY} = \frac{\sum_{t=1}^{T}(X_t - \bar{X})(Y_t - \bar{Y})}{\sqrt{\sum_{t=1}^{T}(X_t - \bar{X})^2 \sum_{t=1}^{T}(Y_t - \bar{Y})^2}}, \quad r_{XY} \in [-1,\, 1]. \tag{5}$$

In Eq. (5), the numerator computes the sample covariance of X and Y the mean product of their respective deviations from sample means $\bar{X}$ and $\bar{Y}$ while the denominator normalises by the geometric mean of their individual sample standard deviations, ensuring that r_{XY} remains bounded on $[-1, 1]$ regardless of the original measurement units or scales.

4.4 K-Means Clustering for System Segmentation

Energy datasets spanning multiple countries or regions exhibit substantial cross-sectional heterogeneity. Some systems combine high installed capacity with low utilisation efficiency; others display rapid growth in renewable share alongside limited grid integration; still others possess mature, stable generation profiles with minimal capacity expansion. Fitting a single diagnostic model uniformly across such structurally diverse data risks averaging over distinct sub-populations, producing conclusions that are descriptively accurate in aggregate but misleading at the level of individual system types.

K-means clustering resolves this by partitioning the n observations—each represented as a feature vector $\mathbf{x} \in \mathbb{R}^d$ comprising variables such as CF_t, RS_t, and Growth_t—into k disjoint clusters $\{C_1, C_2, \ldots, C_k\}$ that maximise within-cluster homogeneity. This is achieved by minimising the total within-cluster sum of squared Euclidean distances from each observation to its cluster centroid:

$$\min_{\{C_i\}_{i=1}^{k}} \sum_{i=1}^{k} \sum_{\mathbf{x} \in C_i} \left\|\mathbf{x} - \boldsymbol{\mu}_i\right\|^2, \tag{6}$$

where $\boldsymbol{\mu}_i = |C_i|^{-1} \sum_{\mathbf{x} \in C_i} \mathbf{x}$ is the centroid of cluster C_i, computed as the arithmetic mean of all assigned observations. The objective in Eq. (6) is non-convex and computationally intractable in the general case; in practice, it is solved iteratively via the Lloyd–Forgy algorithm, alternating between an assignment step—in which each observation is associated with the nearest centroid—and an update step—in which centroids are recomputed as the means of their current membership. Convergence to a local minimum is guaranteed, but the outcome is sensitive to initialisation; K-means++ seeding is therefore employed to place initial centroids in a well-dispersed configuration that reduces the risk of converging to poor local optima. The cluster count k is treated as a hyperparameter and selected using the elbow criterion, which identifies the point beyond which additional clusters yield diminishing reductions in total within-cluster inertia $W(k) = \sum_{i=1}^{k} \sum_{\mathbf{x}\in C_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2$.

Algorithm 2 Diagnostic Analytics Pipeline

Require: Capacity data $\{C_t\}$, actual generation $\{G_t^{\text{act}}\}$, variable pairs (X_t, Y_t), lag range $K_{\max}$, feature matrix $\mathbf{X} \in \mathbb{R}^{n\times d}$, cluster range $[k_{\min}, k_{\max}]$
Ensure: Capacity factors $\{CF_t\}$, correlation matrix $\mathbf{R}$, optimal lag k^*, cluster labels $\mathbf{z}$

Compute CF_t for all t via Eq. (4)
Construct full pairwise correlation matrix $\mathbf{R}$ via Eq. (5)
for $k = 1$ **to** $K_{\max}$ **do** ▷ Lag selection
    Fit OLS model $Y_t = \alpha + \beta X_{t-k} + \varepsilon_t$
    Compute $\text{AIC}_c(k)$ via Eq. (**??**)
end for
$k^* \leftarrow \arg\min_k \ \text{AIC}_c(k)$
for $k = k_{\min}$ **to** $k_{\max}$ **do** ▷ Elbow analysis
    Initialise centroids $\{\boldsymbol{\mu}_i\}_{i=1}^{k}$ via K-means++
    repeat
        Assignment: $z_j \leftarrow \arg\min_i \|\mathbf{x}_j - \boldsymbol{\mu}_i\|^2$ for all j
        Update: $\boldsymbol{\mu}_i \leftarrow |C_i|^{-1} \sum_{j:\, z_j=i} \mathbf{x}_j$ for all i
    until $\max_i \|\boldsymbol{\mu}_i^{\text{new}} - \boldsymbol{\mu}_i^{\text{old}}\| < \delta_{\text{tol}}$
    Record $W(k) \leftarrow \sum_i \sum_{j:\, z_j=i} \|\mathbf{x}_j - \boldsymbol{\mu}_i\|^2$
end for
$k^*_{\text{clust}} \leftarrow$ elbow point of $\{W(k)\}$
Re-run K-means with $k = k^*_{\text{clust}}$; record labels $\mathbf{z}$
return $\{CF_t\}$, $\mathbf{R}$, k^*, $\mathbf{z}$

5 Predictive Analytics: Forecasting and Probabilistic Modelling

5.1 Overview

After we characterised the historical profile of the energy system through descriptive and diagnostic analysis and identified the structural componets that is responsible for the causal mechanisms through diagnostic analysis , Now we move on to the third most important layer which is the predictive analytics . This layer is responsible for addressing the question of , which is likely to occur bascially trying to predict whats going to happen and when , bascially analysing the structure enabling to understand the future evolution that is actually able to quantify and project the probability of achieving specific transition milestone . Predictive analysis where we use to gather all the previous information and try to solve or find some solution to the problem .

The predicted analytics is not just an exploratory data analytics sanity test , It will use statistical inference problem and empirical pattern guidelines and causal structures identified in prior stages that are formalised into a probablistic models that are capabale of genrating a well calibrated custom forecasts with quantified uncertainity

Two distinct but complementary modelling paradigms are employed. The first concerns time series forecasting that given the historical trajectory of a continuous energy variable, the objective is to produce h stepahead point forecasts together with prediction intervals that bound the range of plausible future realisations. The second concerns probabilistic classification: given a vector of observable system characteristics, the objective is to estimate the probability that a country or region will achieve a binary transition outcome such as surpassing a critical renewable energy share threshold within a specified planning horizon. Together, these paradigms provide both the trajectory and likelihood dimensions of the system's future, yielding a complete probabilistic characterisation of the energy transition pathway.

5.2 Linear Trend Model

The linear trend specification provides an analytically tractable and interpretable baseline against which complex models can be benchmarked. In energy systems characterised by sustained longrun expansion of renewable capacity or generation driven by technology cost reductions, policy mandates, and growing climate commitments a deterministic linear trend often accounts for a substantial share of the total variance in the series.

The linear trend model expresses the outcome variable as a deterministic linear function of time plus a stochastic disturbance:

$$Y_t = \beta_0 + \beta_1 t + \varepsilon_t, \tag{7}$$

where β_0 is the intercept representing the fitted value at $t = 0$, β_1 is the slope quantifying the average change in Y per unit of time, and ε_t is the error term. The OLS estimators are obtained by minimising the residual sum of squares RSS $= \sum_{t=1}^{T} \varepsilon_t^2 = \sum_{t=1}^{T}(Y_t - \beta_0 - \beta_1 t)^2$, yielding the closed-form expressions:

$$\hat{\beta}_1 = \frac{\displaystyle\sum_{t=1}^{T}(t-\bar{t})(Y_t-\bar{Y})}{\displaystyle\sum_{t=1}^{T}(t-\bar{t})^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{t}, \tag{8}$$

where $\bar{t} = T^{-1}\sum_{t=1}^{T} t$ and $\bar{Y} = T^{-1}\sum_{t=1}^{T} Y_t$ are the respective sample means. Under the Gauss Markov conditions, the OLS estimator is the best linear unbiased estimator (BLUE), guaranteeing minimum variance among all linear unbiased estimators when ε_t is homoscedastic and serially uncorrelated.

5.3 ARIMA Forecasting

The ARIMA(p, d, q) model is governed by three integer hyperparameters: p, the autoregressive order; d, the degree of differencing applied to achieve stationarity; and q, the moving average order. Using the backshift operator B defined by $BY_t = Y_{t-1}$, the complete model is expressed compactly as:

$$\varphi(B)\,(1-B)^d Y_t = \delta + \theta(B)\,\varepsilon_t, \tag{9}$$

The δ is a constant drift term, and the autoregressive and moving average polynomials are

$$\varphi(B) = 1 - \varphi_1 B - \varphi_2 B^2 - \cdots - \varphi_p B^p, \tag{10}$$
$$\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q. \tag{11}$$

The operator $(1-B)^d$ will applies d and fold differencing to the raw series for $d = 1$, this yields the first difference $\Delta Y_t = Y_t - Y_{t-1}$; for $d = 2$, the second difference $\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}$; and so on. Differencing is applied until the augmented Dickey Fuller (ADF) test fails to reject the null hypothesis of a unit root at the 5% significance level, confirming that stationarity has been achieved. The model parameters $\{\varphi_i\}$ and $\{\theta_j\}$ are estimated by numerically maximising the conditional log-likelihood. The hyperparameter triple (p, d, q) is selected across a candidate grid by minimising:

$$\mathrm{AIC}_c = -2\ln\hat{\mathcal{L}} + 2m + \frac{2m(m+1)}{T-m-1}, \tag{12}$$

where $m = p + q + \mathbf{1}[\delta \neq 0]$ is the total count of estimated parameters. The h-step-ahead point forecast $\hat{Y}_{T+h}$ and its 95% prediction interval are given by:

$$\hat{Y}_{T+h} \pm 1.96\,\hat{\sigma}_\varepsilon \sqrt{1 + \sum_{j=1}^{h-1} \psi_j^2}\,, \tag{13}$$

where $\hat{\sigma}_\varepsilon^2 = (T-m)^{-1}\sum_t \hat{\varepsilon}_t^2$ is the estimated error variance and $\{\psi_j\}$ are the $\mathrm{MA}(\infty)$ representation coefficients computed recursively from $\hat{\varphi}$ and $\hat{\theta}$.

5.4 Logistic Regression for Transition Classification

Logistic regression is well suited to this task: it is a discriminative probabilistic classifier that directly models the conditional probability of a binary outcome from a feature vector, yields interpretable coefficient estimates, and avoids the distributional assumptions on the predictors required by generative alternatives such as linear discriminant analysis.

Let $Y_j \in \{0, 1\}$ indicate whether observation unit j achieves the transition target ($Y_j = 1$) or does not ($Y_j = 0$). The conditional probability is modelled as:

$$P(Y_j = 1 \mid \mathbf{x}_j) = \sigma\Big(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_j\Big) = \frac{1}{1 + \exp\big(-\big(\beta_0 + \sum_{i=1}^n \beta_i x_{ji}\big)\big)}, \tag{14}$$

where $\sigma(z) = (1+e^{-z})^{-1}$ is the sigmoid function mapping the linear predictor onto $[0, 1]$; $\mathbf{x}_j = (x_{j1}, \ldots, x_{jn})^\top$ is the feature vector comprising system indicators such as RS_t, Growth_t, and CF_t; and $\{\beta_i\}$ are the log odds coefficients whose exponentiated values e^{β_i} represent odds ratios. Parameters are estimated by maximising the binary cross entropy log likelihood:

$$\ell(\boldsymbol{\beta}) = \sum_{j=1}^{m} \Big[y_j \ln \hat{p}_j + (1 - y_j) \ln(1 - \hat{p}_j) \Big], \tag{15}$$

subject to L_2 regularisation $\lambda \|\boldsymbol{\beta}\|_2^2$ to mitigate overfitting in high dimensional feature spaces, with regularisation strength λ selected by k fold cross validation. Maximisation of Eq. (15) is carried out via gradient ascent, as no closed-form solution exists for logistic regression.

Algorithm 3 Predictive Analytics Pipeline

Require: Series $\{Y_t\}_{t=1}^{T}$, feature matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$, binary labels $\mathbf{y} \in \{0,1\}^m$, ARIMA search grid $(P_{\max}, Q_{\max})$, forecast horizon H, regularisation grid Λ, learning rate η, tolerance τ

Ensure: Linear trend forecasts, ARIMA forecasts $\{\hat{Y}_{T+h}\}_{h=1}^{H}$ with prediction intervals, class probabilities $\{\hat{p}_j\}$

Fit OLS: compute $\hat{\beta}_0, \hat{\beta}_1$ via Eq. (8)
Apply ADF test to $\{Y_t\}$; determine differencing order d
for $p = 0$ **to** $P_{\max}$; $q = 0$ **to** $Q_{\max}$ **do**
    Fit ARIMA(p, d, q) via maximum likelihood
    Record AIC$_c(p, q)$ via Eq. (12)
end for
$(p^*, q^*) \leftarrow \arg\min_{p,q}$ AIC$_c(p, q)$
Compute h-step forecasts $\hat{Y}_{T+h}$ and 95% PIs via Eq. (13)
Select λ^* via k-fold cross-validation over Λ
Initialise $\boldsymbol{\beta} \leftarrow \mathbf{0}$
repeat ▷ Logistic regression via gradient ascent
    $\hat{p}_j \leftarrow \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_j)$ for all j
    $\nabla \ell \leftarrow \mathbf{X}^\top (\mathbf{y} - \hat{\mathbf{p}}) - 2\lambda^* \boldsymbol{\beta}$
    $\boldsymbol{\beta} \leftarrow \boldsymbol{\beta} + \eta \nabla \ell$
until $\|\nabla \ell\|_2 < \tau$
return Linear/ARIMA forecasts and PIs, $\{\hat{p}_j\}$

6 Prescriptive Analytics: Optimisation and Decision Intelligence

6.1 Overview

The prescriptive analytics layer represents the culmination of the proposed framework, translating the forecasts and probabilistic estimates generated in Sect. 5 into concrete and actionable decision strategies. Whereas predictive analytics is concerned with estimating what the future holds so it is mainly designed to predict the future or future enhancement or evolution of a system , prescriptive analytics addresses the normative question of what brought to be done in order to achieve desired outcomes under real world constraints. This distinction is fundamental a decision maker who learns that a country is unlikely to meet its renewable energy target without intervention gains relatively little value from that knowledge alone the framework must additionally specify which interventions are optimal, under what resource constraints, and with what degree of confidence given the inherent uncertainty pervading the system.

The prescriptive layer rests on three methodological pillars. First, constrained optimisation formulated as a linear programme or, in more general settings, as a non-linear programme identifies the allocation of resources across competing investment options that minimises total cost or maximises a specified performance objective while respecting physical, financial, and regulatory constraints.

6.2 Monte Carlo Simulation for Uncertainty Quantification

An investment strategy that is optimal under expected parameter values may perform substantially worse or even become infeasible under adverse realisations. Monte Carlo simulation addresses this limitation by treating uncertain inputs as random variables drawn from specified probability distributions and evaluating the objective across a large ensemble of simulated realisations.

Let $\boldsymbol{\xi} \in \mathbb{R}^q$ denote the vector of stochastic parameters with joint distribution $\boldsymbol{\xi} \sim F$. The expected value of the objective function at $\mathbf{x}^*$ under parametric uncertainty is estimated as:

$$\mathbb{E}_{\boldsymbol{\xi}}[f(\mathbf{x}^*, \boldsymbol{\xi})] \approx \frac{1}{N} \sum_{i=1}^{N} f\left(\mathbf{x}^*, \boldsymbol{\xi}^{(i)}\right), \quad \boldsymbol{\xi}^{(i)} \overset{\text{i.i.d.}}{\sim} F, \tag{16}$$

where N is the number of simulation trials. By the strong law of large numbers, the estimator in Eq. (16) converges almost surely to the true expectation as $N \to \infty$. The associated Monte Carlo standard error $\mathrm{SE} = \hat{\sigma}/\sqrt{N}$ where $\hat{\sigma}^2$ is the sample variance of $\{f_i\}_{i=1}^N$ quantifies estimation precision, and a $(1-\alpha)$ confidence interval on the expected objective is:

$$\mathbb{E}[f] \in \left[\bar{f} - z_{\alpha/2}\frac{\hat{\sigma}}{\sqrt{N}},\ \bar{f} + z_{\alpha/2}\frac{\hat{\sigma}}{\sqrt{N}}\right], \tag{17}$$

where $\bar{f} = N^{-1}\sum_{i=1}^{N} f_i$ and $z_{\alpha/2}$ is the standard normal critical value (e.g., $z_{0.025} = 1.96$ for a 95% interval). Beyond point estimation, the simulation produces the full empirical distribution of f, enabling computation of risk measures such as the Value at Risk and Conditional Value at Risk, which quantify tail exposure in the investment strategy.

6.3 Rule-Based Decision Support

Quantitative optimisation and simulation provide the numerical backbone of the prescriptive layer, but decision-makers in the energy sector also rely on domain expertise and qualitative judgement that cannot always be encoded within a mathematical programme. Rule-based decision support bridges this gap by representing expert knowledge as a structured collection of condition action pairs $\mathcal{R} = \{(c_r, a_r)\}_{r=1}^{R}$, where each condition c_r is a logical predicate over the system state variables produced by prior analytical layers, and each action a_r is a targeted intervention recommendation. These rules are evaluated after Monte Carlo simulation to ensure their conclusions remain robust to parametric uncertainty. Representative rules include:

- **R1** (Utilisation Gap): If $CF_t < 0.30$ and $C_t > C_{\text{ref}}$, then prioritise grid modernisation and transmission expansion to unlock latent generation capacity.
- **R2** (Substitution Barrier): If $r_{XY} < 0.20$ between renewable capacity additions and fossil fuel generation reductions, then flag a structural substitution failure and initiate a grid flexibility and battery storage deployment study.
- **R3** (Transition Readiness): If $P(Y = 1 \mid \mathbf{x}) > 0.75$, then classify the region as transition ready and recommend accelerated green financing and procurement policy.

The rule based system is designed and implemented to be more transparent , auditable extensible and extendable ensuring that the prespective layer remains interpretable to non technical stakeholder while preserving the full rigour of the underlying quantitive analysis

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Algorithm 4 Prescriptive Analytics Pipeline

Require: Cost vector $\mathbf{c}$, constraint matrix A, bound vector $\mathbf{b}$, stochastic distribution F, MC trials N, perturbation δ, rule base $\mathcal{R}$, diagnostic outputs $\{CF_t\}$, $\mathbf{R}$, predictive outputs $\{\hat{p}_j\}$
Ensure: Optimal allocation $\mathbf{x}^*$, expected cost $\bar{f}$, 95% CI, sensitivity indices $\{S_k\}$, ranked recommendations

Solve LP (**??**) via simplex or interior-point $\to \mathbf{x}^*$
for $i = 1$ **to** N **do** ▷ Monte Carlo uncertainty propagation
    Draw $\boldsymbol{\xi}^{(i)} \sim F$
    Evaluate $f_i \leftarrow f(\mathbf{x}^*, \boldsymbol{\xi}^{(i)})$
end for
$\bar{f} \leftarrow N^{-1} \sum_{i=1}^{N} f_i$
$\hat{\sigma} \leftarrow \sqrt{(N-1)^{-1} \sum_{i=1}^{N} (f_i - \bar{f})^2}$
Construct 95% CI via Eq. (17)
for $k = 1$ **to** q **do** ▷ Parameter sensitivity ranking
    $S_k \leftarrow \left[f(\mathbf{x}^*, \boldsymbol{\xi} + \delta \mathbf{e}_k) - f(\mathbf{x}^*, \boldsymbol{\xi})\right]/\delta$ via Eq. (**??**)
end for
Rank $\{|S_k|\}$ in descending order; flag top-m as policy-critical
for each rule $(c_r, a_r) \in \mathcal{R}$ **do** ▷ Rule-based inference
    Evaluate c_r against $\{CF_t\}$, $\mathbf{R}$, $\{\hat{p}_j\}$
    if c_r is satisfied **then**
        Append a_r to ranked recommendation list
    end if
end for
return $\mathbf{x}^*$, $\bar{f}$, 95% CI, $\{S_k\}$, recommendation list

---

7 Dataset Description and Preprocessing Framework

7.1 Data Sources and Construction

The data has been obtained ethically from globally recognized energy and sustainability dataset repositories , this are selected to ensure both reliable and analytically deep which enables to gain some extra insights which would be a key important factor for choosing the right database for the project , The repostories provide has standard indicators on electricity generation , installed capacity , actual capacity , energy type , energy production and other reports that is well aligned with the SDG sustainable Development goal indicators there are some additional dataset included for cross referencing which include macro level characteristics such as fossil fuels dependency and overall energy consumption patterns

The source mainly obtained from IRENA , WHO and World Data Bank which specifically aligns our SDG goal includes details like capacity generation factors and other important factors The combined dataset spans around 2000 to 2023 enabling for longitudinal examination ,time series analysis and forecasting trend analysis etc to understand change in system and global renewable energy resources and its infrastructure it also contain country level information containing annual observation of installed capacity(MW), electricity generation (GWh) and energy mix composition . This dataset servers as the foundation for descriptive diagnostic and time series analyses also predictive analytics and machine learning

7.2 Dataset Characteristics and Statistical Structure

The datasets included both temporal and cross sectional dimensions The temporal components actually deals with the annual observation over a 24 - year period while the cross sectional dimension actually captures multiple countries with heterogeneous energy system and transition pathways. Each observation corresponds to a country year pair forming a multi index dataset that is suitable for both time series and cross sectional analysis

The main feature space includes installed capacity that describes the overall installed capacity that each technology or renewable power plant can theoretically hold and produce , electricity generation (GWh) how much electricity is actual transferring through out the power grid , other features like renewable energy share per captia , fossil fuel share , fossil fuel consumption per country and other technology specific factors like solar wind generations and additionaly adding up with performance and effiencey metrics included with system level behaviour

predictive modeling uses binary target variable is defined below

$$Clean\_Energy\_Dominant = \begin{cases} 1, & \text{if renewable share exceeds threshold (Clean energy dominat) } \tau \\ 0, & \text{otherwise (not deominat fossil fuel dependent)} \end{cases} \quad (18)$$

where τ represents a policy relevant threshold used to distinguish between clean energy dominant and fossil dependent systems.

The dataset is Multi dimensional , heterogeneous reflecting variations across cross continent geographically technologically and time based structure , this enables over application to analyze diverse paradigms and use statistical inferencing time series models and even supervised machine learning approaches

7.3 Data Preprocessing and Transformation Pipeline

This process ensure for preprocessing is implemented to make sure the data is stable , has defined structure , integrity before going for analysis and downstream tasks

7.3.1 Data Cleaning and Consistency Enforcement

The raw data pulled from many sources will go rigorous cleaning procedure to ensure the structural consistency , this process will handle missing values interpolation techniques for time series data and also ensure structural and schema level validation

7.3.2 Unit Standardization and Normalization

Our analysis contains many meaningful units and measure that are highly crucial and need to be standardized , like installed capacity in megawatts , electricity generation in gigawatts , this need to span around globally in different time periods need to normalize and bring standard convention for easy globally identifiable measuring analysis

7.3.3 Temporal Alignment and Indexing

Since we may be dealing with multiple data sources , we are trying to adjust the temporal alignment , unified time series indexing to synchronize all the variables at a yearly resolutions

7.3.4 Feature Engineering and Derived Metrics

we need to construct several features including growth dynamics are captured using rate of change

$$Growth_t = \frac{F_t - F_{t-1}}{F_{t-1}} \tag{19}$$

The efficiency of a system is quantified using capacity factor

$$Capacity\ Factor = \frac{Actual\ Generation}{Installed\ Capacity \times 8.76} \times 100 \tag{20}$$

renewable resource contribution measured through

$$Renewable\ Share = \frac{Renewable\ Generation}{Total\ Generation} \tag{21}$$

the output gap

$$Output\ Gap = Expected\ Generation - Actual\ Generation \tag{22}$$

The theoretical maximum based on the installed capacity

7.3.5 Data Transformation and Encoding

The dataset need to prepared for analytical and predictive modeling categorical variables such as country , region and technology type are encoded into numerical representations

7.3.6 Outlier Detection and Treatment

The outliers are identified using statistcal methods like Z-score analysis or IQR interquartile range . Extreme values are need to handled correctly to prevent distortion

7.3.7 Model-Ready Dataset Preparation

The dataset is further structured into a feature matrix X and a target vector y. The feature scaling is applied necessary this enables us to apply predictive models and machine learning such downstream task using our dataset

8 System Architecture

This section is dedicated to our proposed architectural design of our analytics system which is designed based on scalability , system design principles , isolated de coupled modules system with stable communication protocols and storage integrations in mind

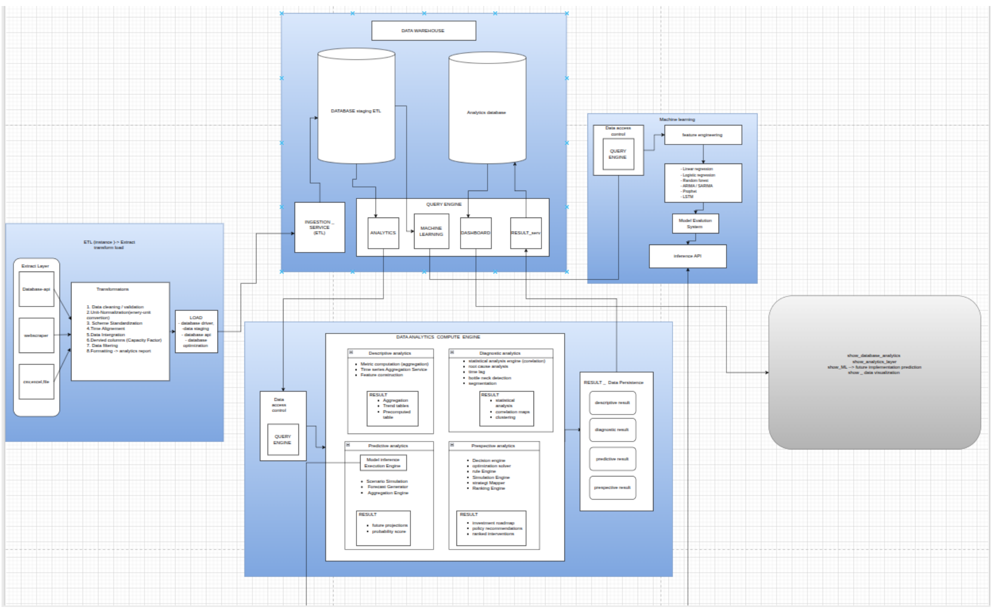


Fig. 1 Complete system architecture.

The System consists of de coupled isolated modules integrated like each module has a specific task or services that they initiate starting from , ETL layer (Extract Transform layer) act as a Data source Layer , Data ingestion layer , data processing layer , data transformation layer , The we have a data warehouse service which actually stores the processed data and analytics data these two storage containers are inside a warehouse service which acts a medium to store retrieve processed information that can be handled by business and other third party service api for their analytics and machine learning prospects also we need make sure it also curated for our SDG analytics process and our predictive machine learning services Data warehouse also known as Business layer contains all the business information to make useful insights .After the warehouse process have been initiated now for flow moves to a very important structure by far the analytics layer , which is responsible for data analytics deriving insight and predictions , the data analytics compute engine consists of descriptive analytics , diagnostic analytics , predictive analytics and perspective analytics then next parallel layer is machine learning layer this is actually designed for prediction and also part of the predictive analytics where it uses machine learning API to preprocessed and perform the predictive analysis Machine learning layer actually uses processed data from the warehouse just like the analytics layer this help to get us high accuracy and high precision for our future predictions and insight , The final layer is dashboard layer where the clients and stakeholder meant to see the finished process data in a very well designed high quality graphical user interface format .

8.1 ETL Architecture

The ETL Extract Load Transform , performs , data extraction , data transformation , data cleaning normalization and , data staging layer .

8.1.1 Extract Layer

the ETL starts from the extract layer where it extract data from different data sources and endpoints , these may include like CSV , web application api , third party api , web scrapping services and further more the extraction layer extract data from all of this resource and get ready to transfer to the next layer , the extraction which actually the base starting point of ingestion layer in a analytics pipeline

8.1.2 Transformation Layer

The Transformation layer , responsible for cleaning validating unit normalization fixing time series alignment and generating derived columns, data filtering, data and formatting also scheme standardization , once the transformation is done now how data is ready for next layer which the data staging or load phase.

8.1.3 Load Layer

The Load phase , usually have take in the transformed and cleaned dataset , will load the data into drivers or api that directly connect to the data warehouse serves or repositories

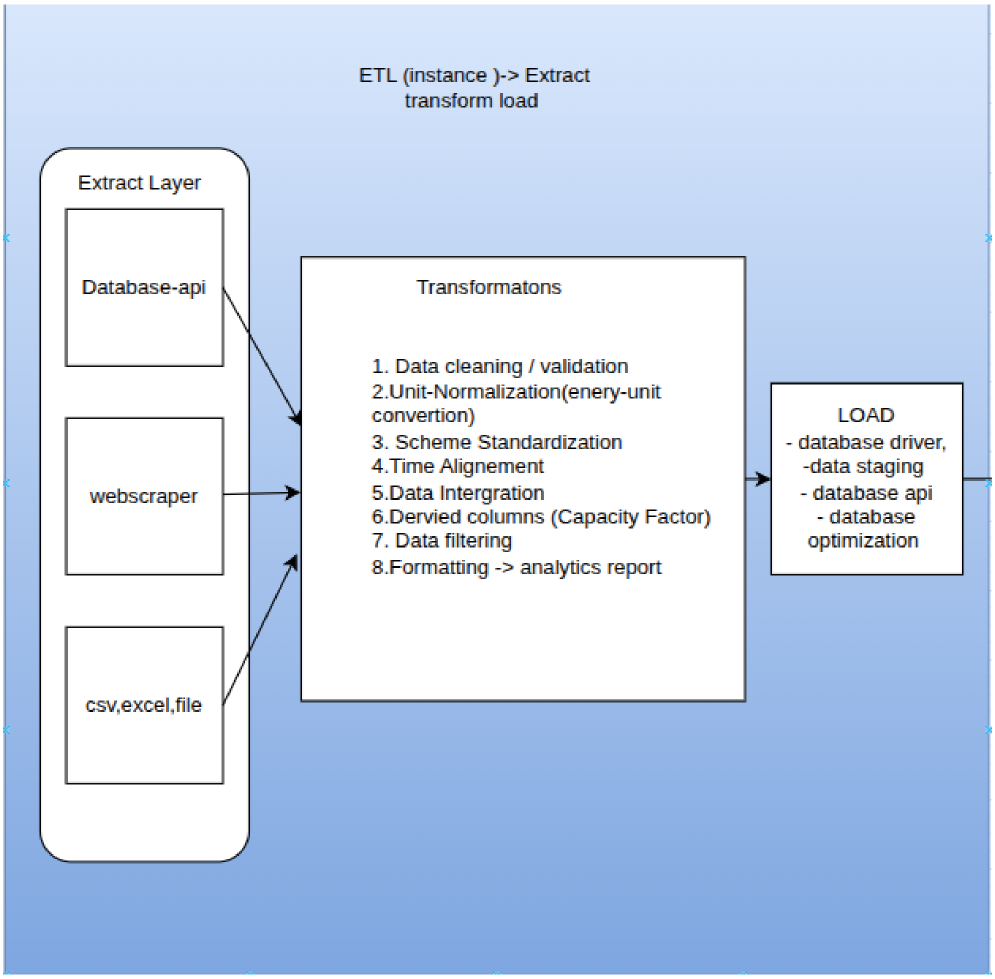


Fig. 2 ETL full pipeline architecture.

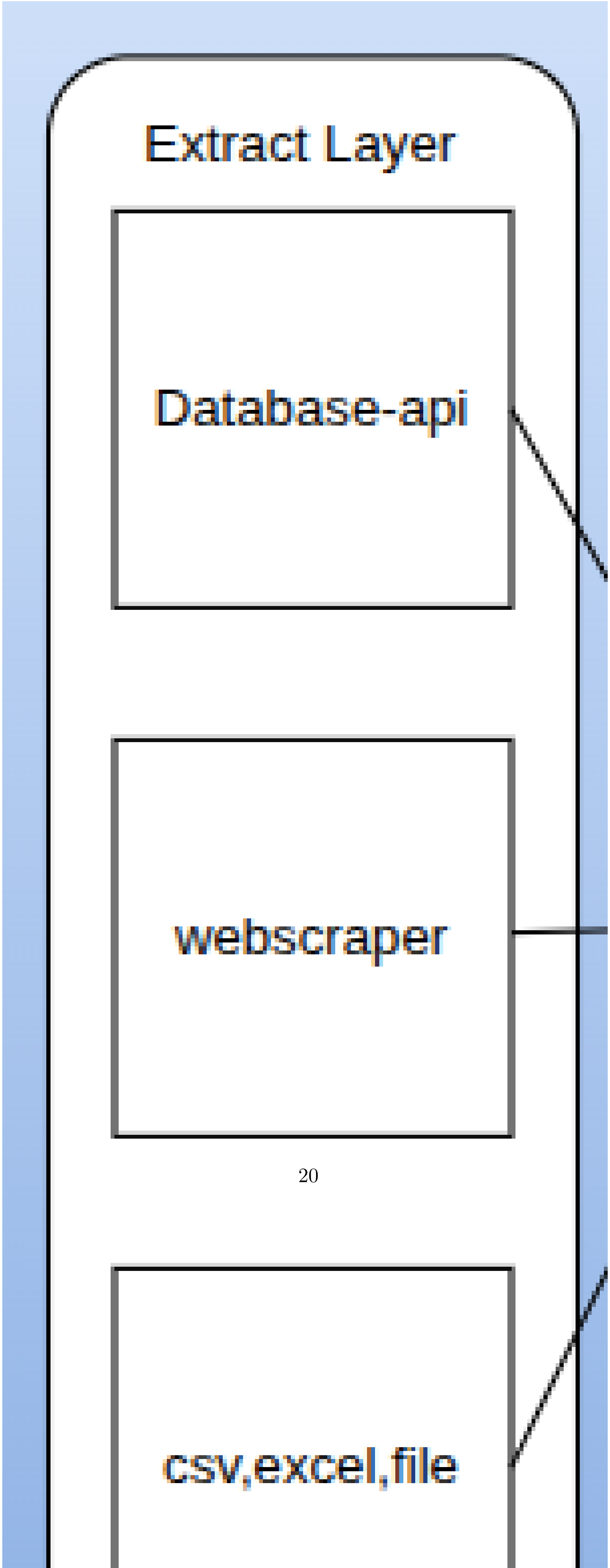
8.2 Data Warehouse Layer

The data warehouse , a centraalised repository storing processed information used to store large amount structured data usually query optimized format so query engines can extract data from it since warehouse contains processed data it is mainly used by data analytics compute clusters and machine learning system

The Data storage warehouse contains

Database staging and ETL layer , this where the stagged pre processed data will inject into the main database inside the data warehouse that can be used later by analytics engine

Analytics database A form of central database inside the warehouse along side with main processed Database , the main reason analytics database exist is because



Extract Layer
Database-api
webscraper
csv,excel,file

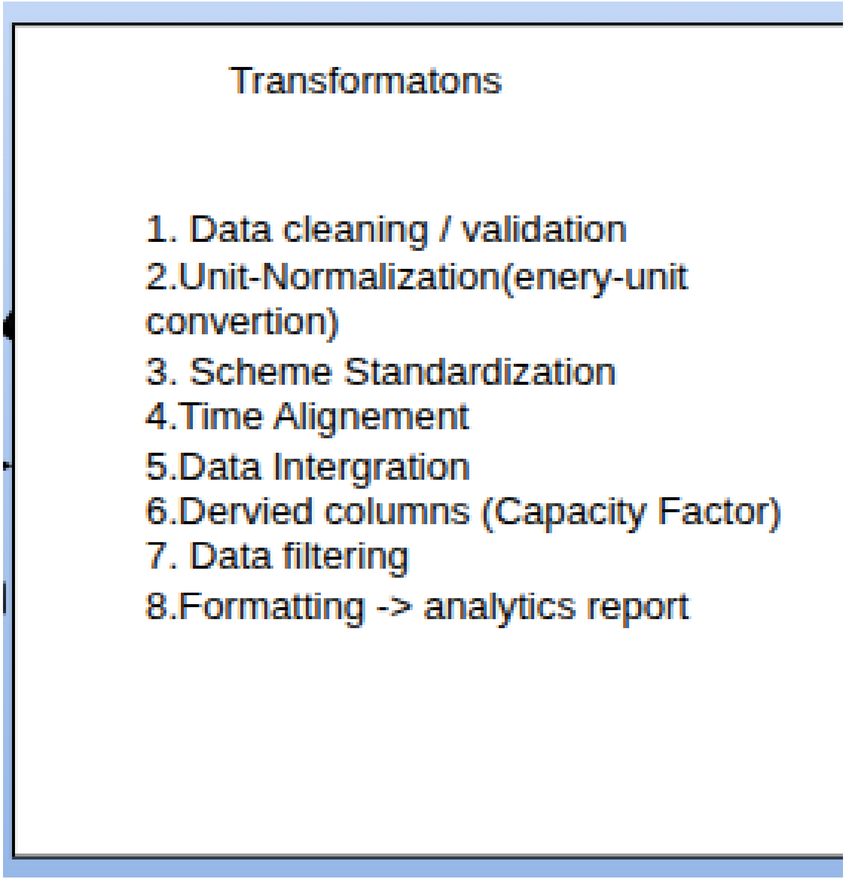


Fig. 4 Transformation layer subsystem.

once you have computed the analytics cluster , the data from the analytics cluster will save inside the analytics database later can be used by frontend by end user report or may be an external analytics engine for find tuning or can be used to verify historical data or data aggregations

Query Engine Designed for to turn raw or processed data into useful operation , the data is obtained from the storage unit and will able to process using sql queries via cron jobs in more tightly integrated automation systems

Dashboard,

The dashboard main frontend is the visual control system for our analytics engine that will display graphs , KPIs summary predictions , individual breakdown on countries , technology trend over time with visual interaction able to analyse the dataset with more visual form etc .

Result services

The Result service is designed to accumalated all the result from the analytics engine and store or serve into different layer like dashboard or analytics engine

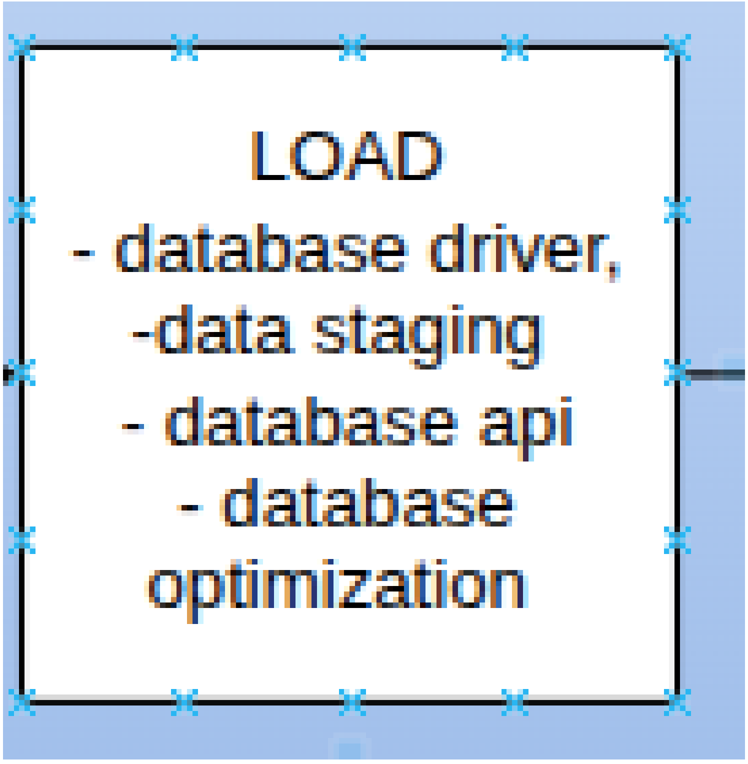


Fig. 5 Load layer subsystem.

8.3 Analytics Compute Engine

The Analytics compute engine instance where the main data analytics operation takes place analytics layer is connect with Data ware house layer where they directly query data with the help query engine and drivers necessary to extract processed information from the database .T he analytics layer this will include the major componets given below "

Descriptive analytics The main starting foundational layer focusing on summarizing historical data and current data into answers , Descriptive analysis mainly answer for what happenend ? , Descriptive analytics will give a clear summary of what was happened into accurate KPI system that enables stake holders and client have easy time interacting with , other componets include metric computation , time series aggregation , feature construction

Diagnostic analytics, The Diagnostic analysis move beyond the classic summary from Descriptive analysis will answer question like why did it happen ? . so in diagnostic analytics it will analyze the underlying causes of observed outcomes by leverging tools like root cause analysis , anomaly detection correlational analysis , Thus enabling us to identify operational inefficiencies system failures and weak strategy

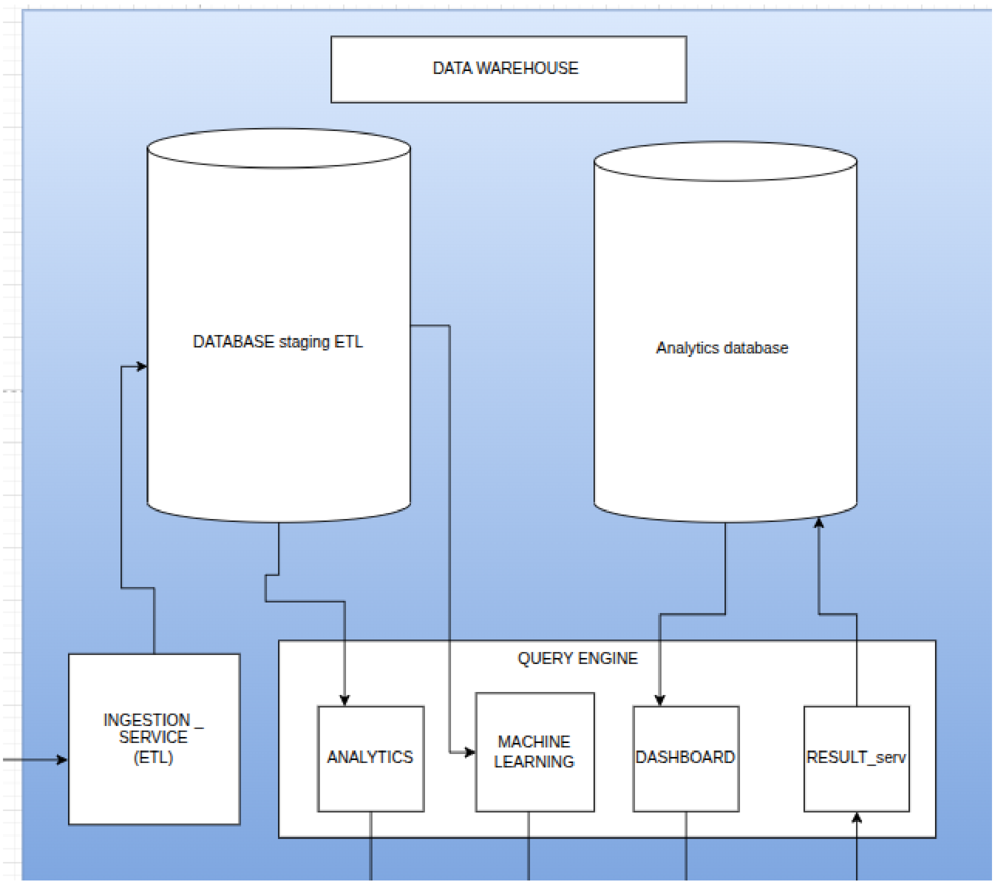


Fig. 6 Data warehouse architecture.

Predictive analytics, The predictive analytics will addresses the question like what is likely to happen ? , by leveraging historical patterns , statistical , machine learning and math models to analysis and predict based on the data , This stage huge hugely driven by machine learning models using classification regression analysis type of problem we are dealing with predictive analysis , we can use time series models like ARIMA prophet for forecasting and predictions analyzing system trends , energy loss over time , predictive maintenance etc

Prescriptive analytics. The prescriptive analysis answer the question such as like what should be done ? This layer focuses on the actionable and well optimized decisions this layer combines results from predictive analysis and their insights to create optimization techniques to improve decision based on the predictions , These involve linear programming , simulation modeling ,reinforcement learning and heuristic approaches

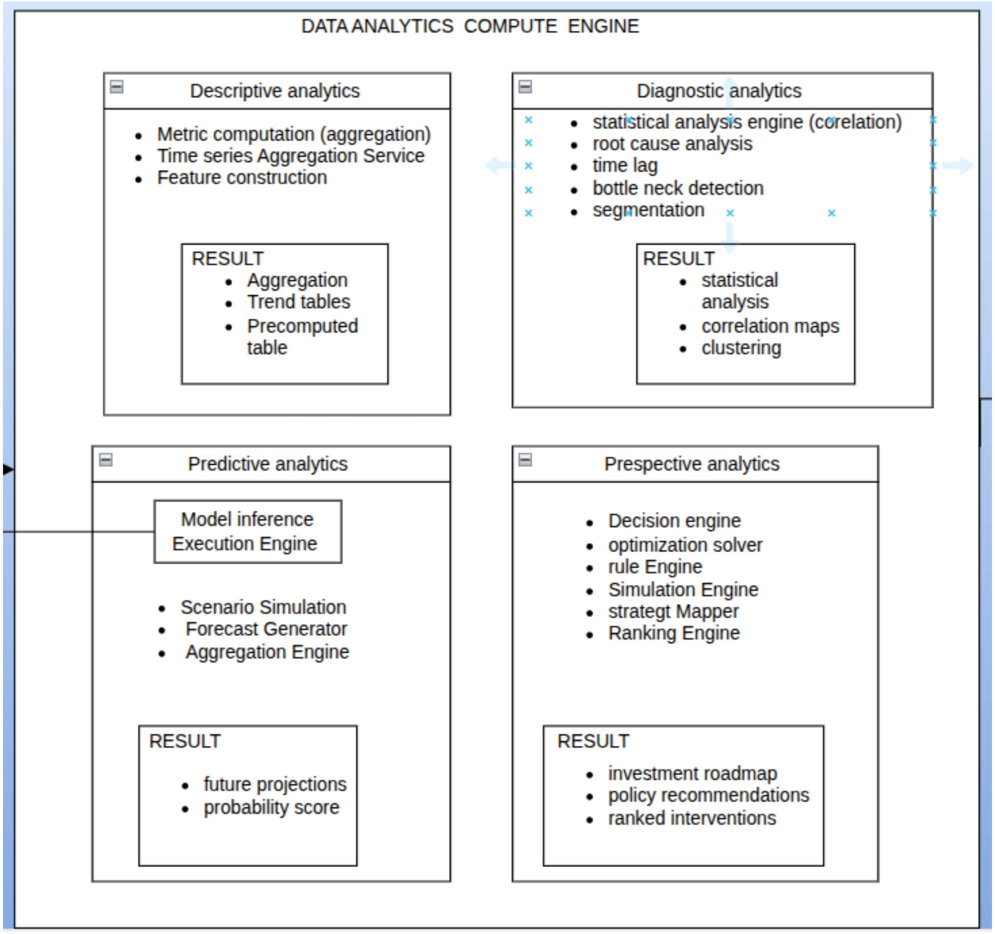


Fig. 7 Analytics compute engine.

8.4 Machine Learning Layer

The Machine learning system is mainly used by the predictive layer which is used inference API to run pretrained models from the machine learning instance , The machine layer extract data from the already processed Data warehouse services

Feature engineering and extraction Turing raw data into meaningful inputs feature vector , can use dimensionality reduction , encoding features like scaling categorical encoding handling missing values this layer is more independent to the machine learning layer doesnt depend on the etl layer , even though data is processed this requires some fine tuning for machine learning models

Regression analysis The most popular statistical analysis tool are linear and logistic regression , linear regression a type of supervised model used for predicting the

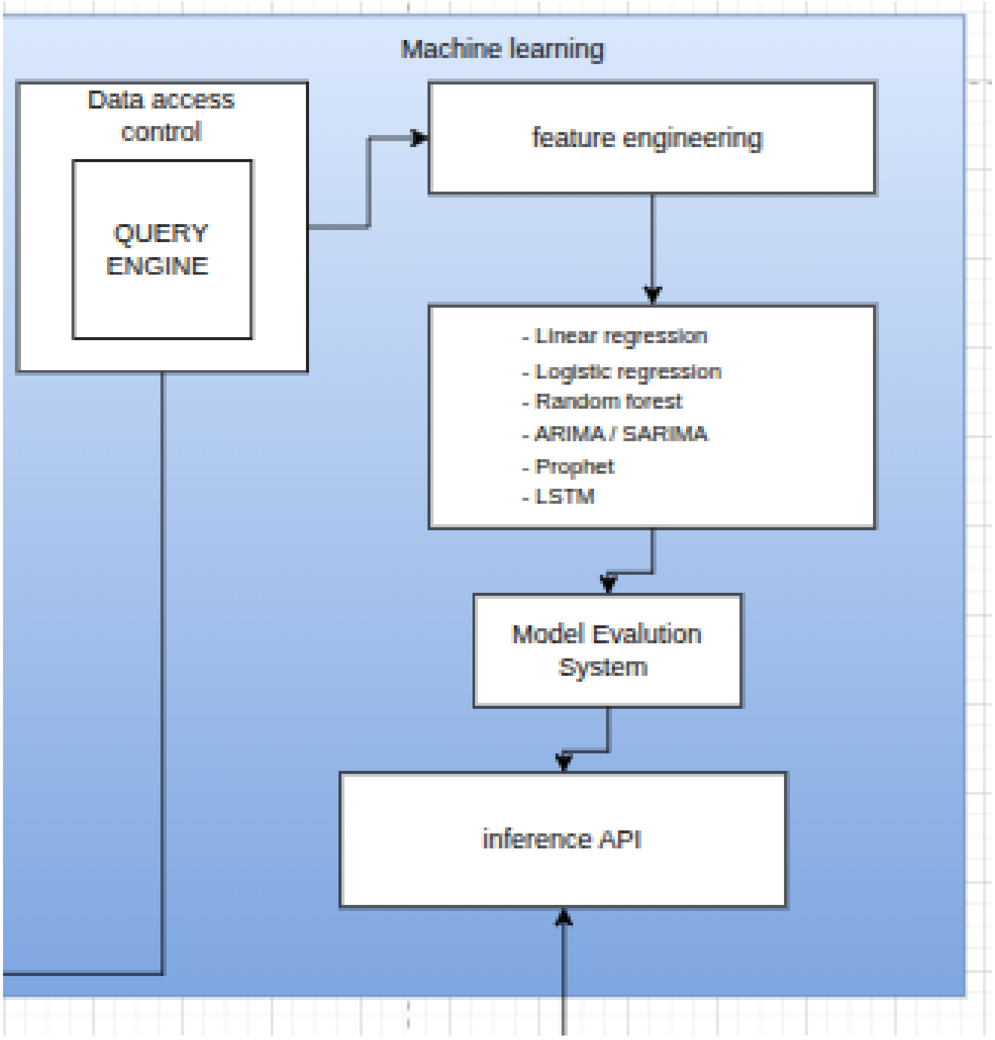


Fig. 8 Machine learning layer.

continuous numerical by modeling a linear relations ship with independent and dependent variables , logistic regression is a classification algorthm used for prediction like yes or no , for example where a country is clean energy domaint or not .

Random Forest A type of decision tree that uses ensemble machine learning methods to improve overall prediction and accuracy to avoid overifting

ARIMA The ARIMA is actually a type of statistical model designed for time series forecasting to predict future based on historical sequential time series data

Model evaluation, evaluatles models using metrics like accuracy precision , recall and RMSE or F1 score

Inference API. A interface usually designed machine learning models to get and outpur response as an API so other componets on the analytics system can actually use the machine learning predictions or use it for inference , the inference api is used by the predictive layer to run pre trained inference and receive output

8.5 Result Persistence Layer

This layer stores:

The Result persistence layer is actually responsible for storing the result from the different analytics services The layer stores data from

Descriptive output - usually aggregation , trend tables and precomputed tables from the descriptive analytics from the data analytics compute engine

Diagnostic layer - returns statistical finding , clustering and correlations maps

predictive analysis output - generates result as future projections and probability score

Descriptive output - output as investment roadmaps , policy recommendations , ranking

The result persistence layer will get all the data from following layers and staged them into analytics layer , later can be used for KPI representation , summarization and predictions

8.6 Visualization Layer

The main visualization and delivery layer that will display graphs, KPIs summary predictions, individual breakdown in countries, technology trend over time with visual interaction that allows analyzing the data set in more visual form, etc.

8.7 Architectural Synthesis

The architecture forms a vertically integrated research-grade computational ecosystem.

The architecture has integrated in a such way that it is de coupled for horizontal integration for future scalability reasons , this isolated integration will allow to set up a distributed computer environment to distributed each components and use communication layer to communicate via each isolate environment or containerized environment to run data on a larger scale

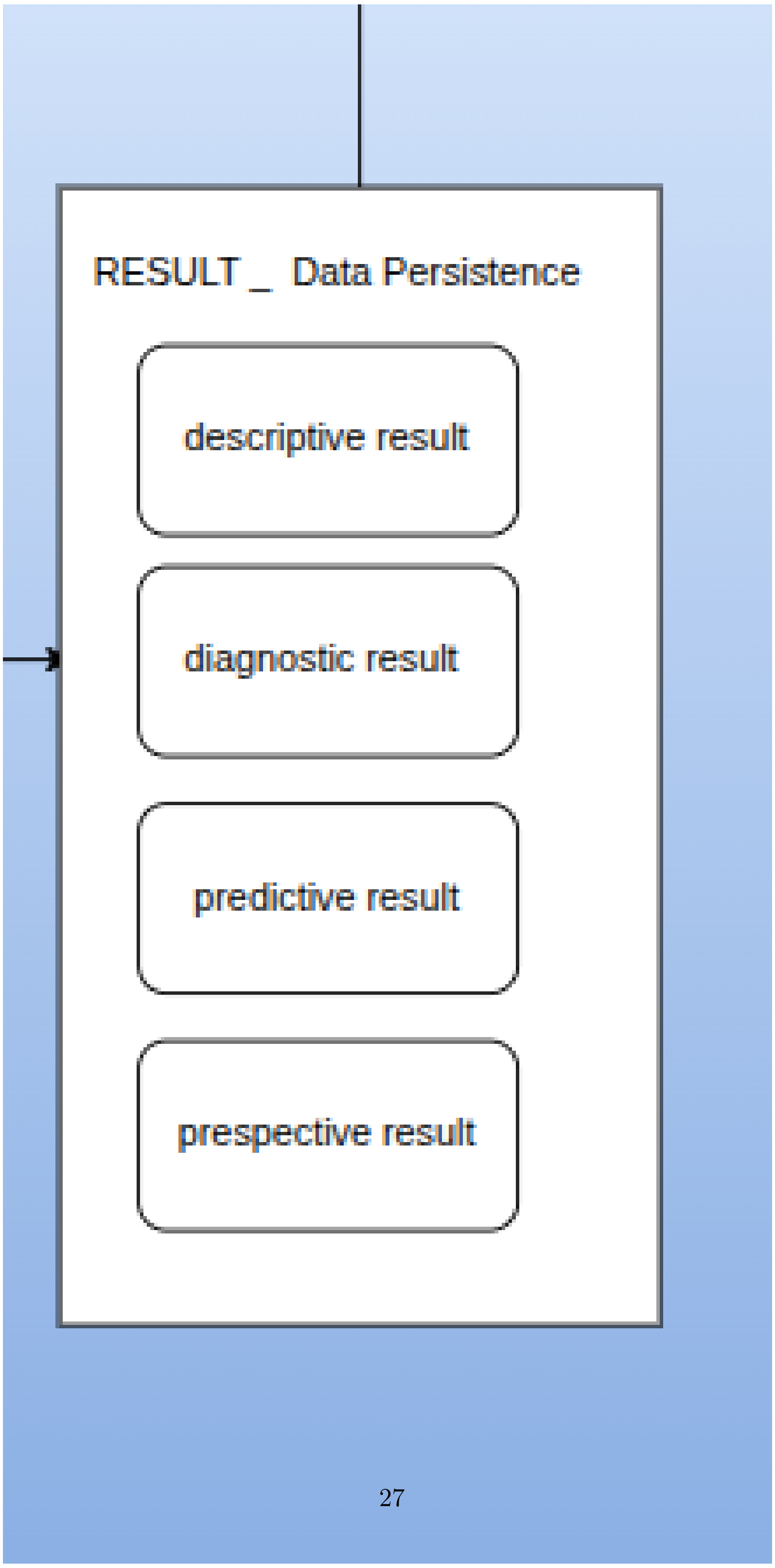


Fig. 9 Result persistence architecture.

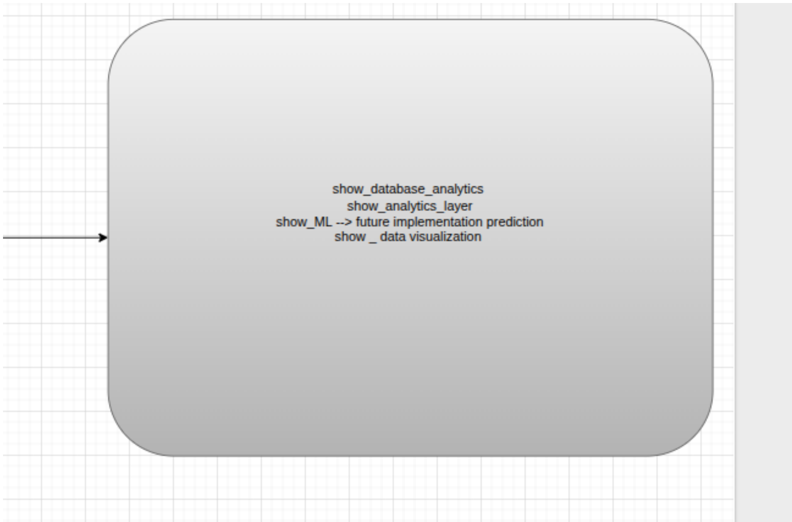


Fig. 10 Visualization and delivery layer.

9 Results

This section is meant to show our empirical findings derived from the multi layer analytical framework capturing global energy systems , behavior efficiency and their transition

9.1 Descriptive Analytics: Global Energy Transition Dynamics

9.1.1 Installed Capacity versus Electricity Generation

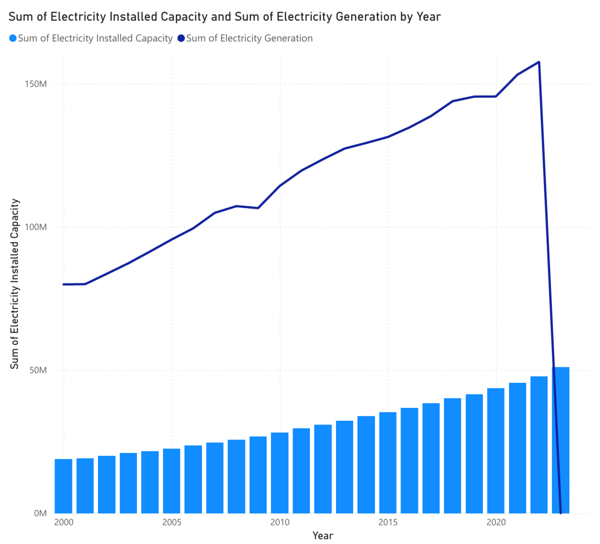


Fig. 11 Global installed electricity capacity and total electricity generation, 2000–2023.

The results has a sustained divergence problem compared to installed capacity realize electricity generation . while capacity expanded significantly but the growth in electricty remained comparatively moderate . This widens the gap and highlight the structural imbalance indicating that increase in capacity do not automatically translate into proportional energy output

9.1.2 Cumulative Technology Performance Efficiency

The assessment of cumulative performance efficiency shows fossil fuel systems maintain by far the highest efficiency (2158.25; 28.00%), followed by hydropower excluding pumped storage (1715.30; 22.26%), bioenergy (1673.77; 21.72%), wind energy (1509.45; 19.59%), and solar energy (649.56; 8.43%).

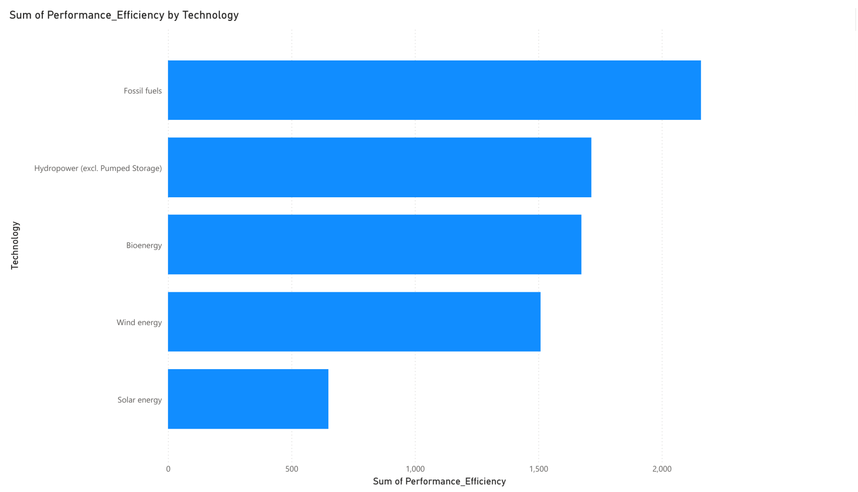


Fig. 12 Cumulative performance efficiency by technology, 2000–2023.

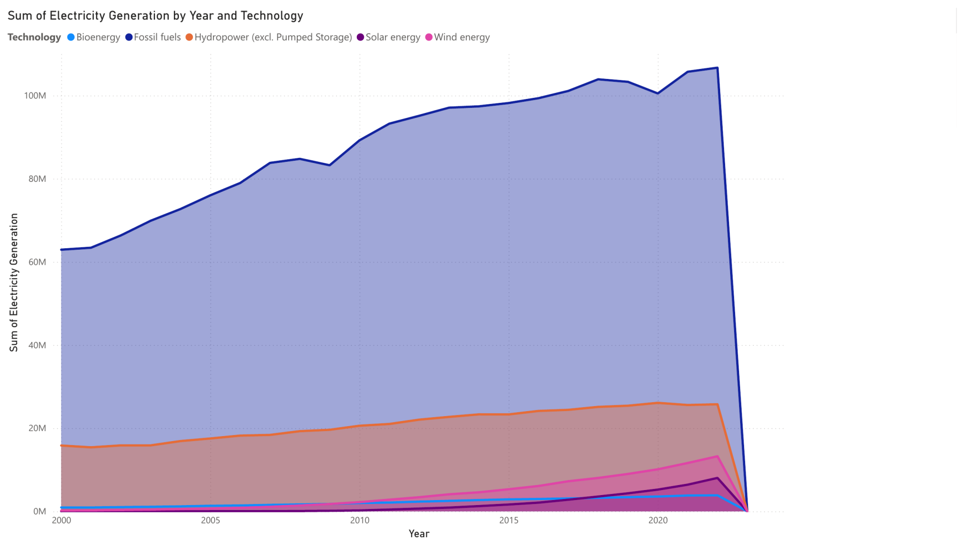


Fig. 13 Global electricity generation by technology, 2000–2023.

9.1.3 Electricity Generation by Technology

Fossil fuel based generation system still remains the dominant player throughout the observation . although wind solar and hydro shows some accelerated growth particular starting from post 2010 phase yet their expansion doesn't result in overall replacement of fossil fuels

9.2 Diagnostic Analytics: Structural Efficiency Gaps

9.2.1 Annual Performance Efficiency Trends

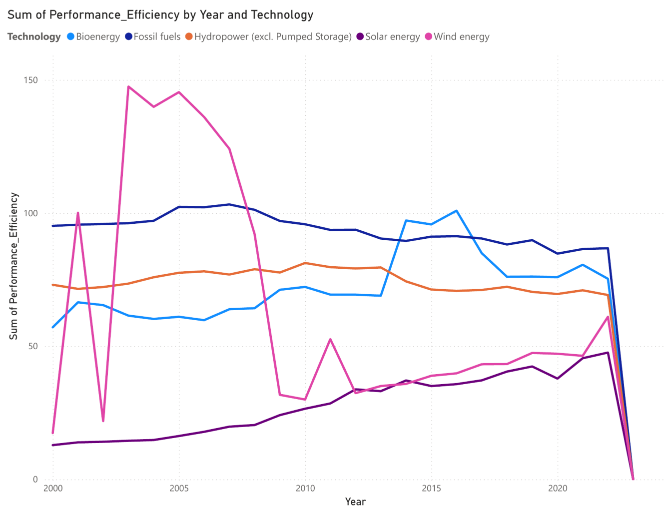


Fig. 14 Annual performance efficiency trends by technology.

Fossil fuel technologies exhibit consistently high and stable efficiency profiles over time. Hydropower and bioenergy maintain moderate yet stable efficiency levels. Wind energy initially displays higher variability, likely reflecting location specific optimisation effects, before stabilising in later periods. Solar energy demonstrates gradual improvement, indicating progressive technological and operational maturation.

9.2.2 System-Wide Efficiency Composition

System wide efficiency patterns show a early sign of peak stabilization at lower levels . This trend may indicates there is a renewable resources penetration increases , There is also a recorded efficiency trade off issue at system level

9.2.3 Correlation and Lag Dynamics

The statistical relationship between renewable capacity growth and fossil generation is weak we can use peasron corelation to identify the coefficient approximately $r \approx 0.12$. The capacity increase indication doesn't directly correspond to immediate reduction in fossil fuel consumption

Lag analysis introduces and approximate delay of three years between capacity expansion and observable impacts on the electricity generation . This is just a temporal

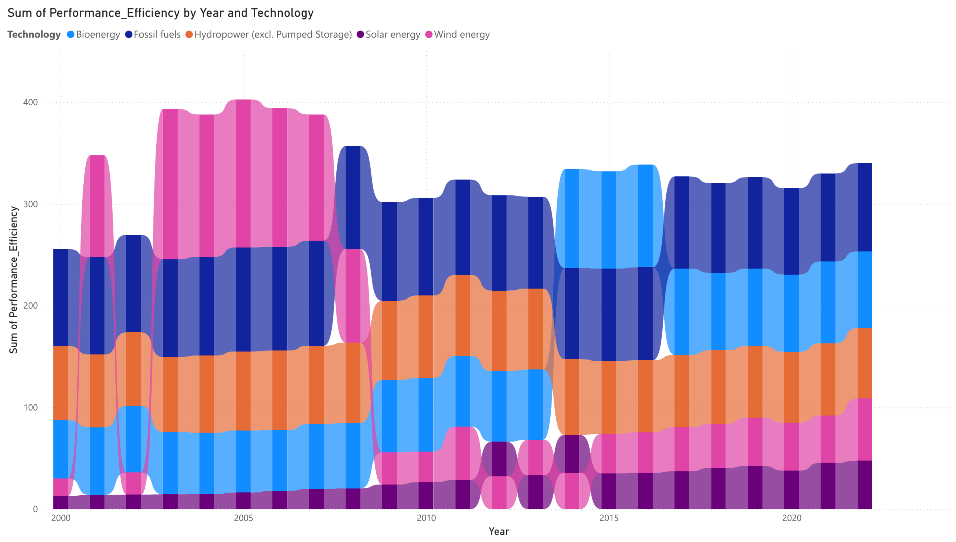


Fig. 15 System-wide stacked annual performance efficiency.

offset that actually reflects the time that is required for the infrastructure integration and grid adaptation and system level optimizations

9.2.4 Regional Structural Divergence

The analysis highlights the significant disparities inconsistency in energy performance , This discrepancies can indicate the infrastructure limitations including grid capacity integration capabilities these play a decisive role in shaping the overall system efficiency

9.3 Predictive Analytics: Forecasting Transition Pathways

The Time series forecasting models such as ARIMA(2,1,1) model achieved a mean absolute percentage error (MAPE) of 2.1%, indicating a high level of predictive accuracy. Projections may suggest that global electricity generation will reach approximately 175 190 million GWh by 2028.

From a technological standpoint, The wind energy is projected to surpass hydropower as the leading renewable electricity source between 2026 and 2027 wind energy has been a efficicent solution for a lot of emerging developing nations such as BRICS . indicating strong discrimination between countries progressing toward clean energy dominance and those constrained by structural limitations.

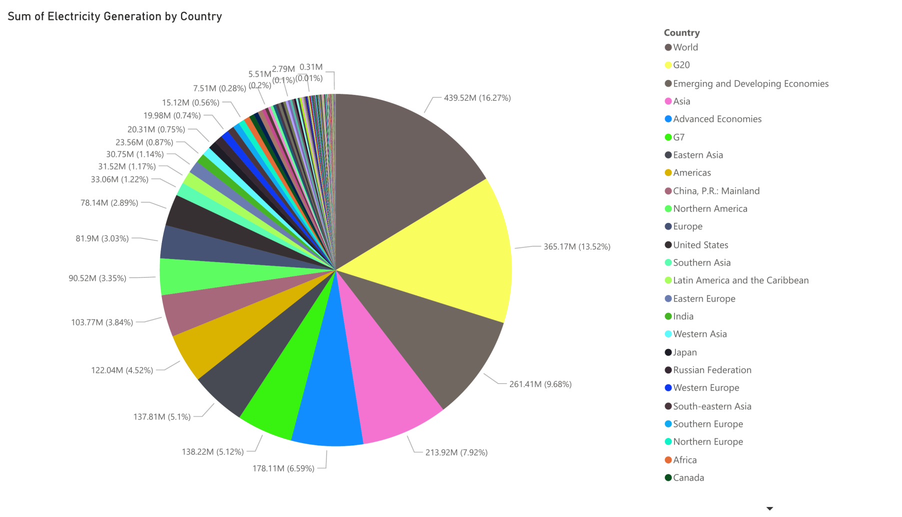


Fig. 16 Regional cumulative electricity generation share.

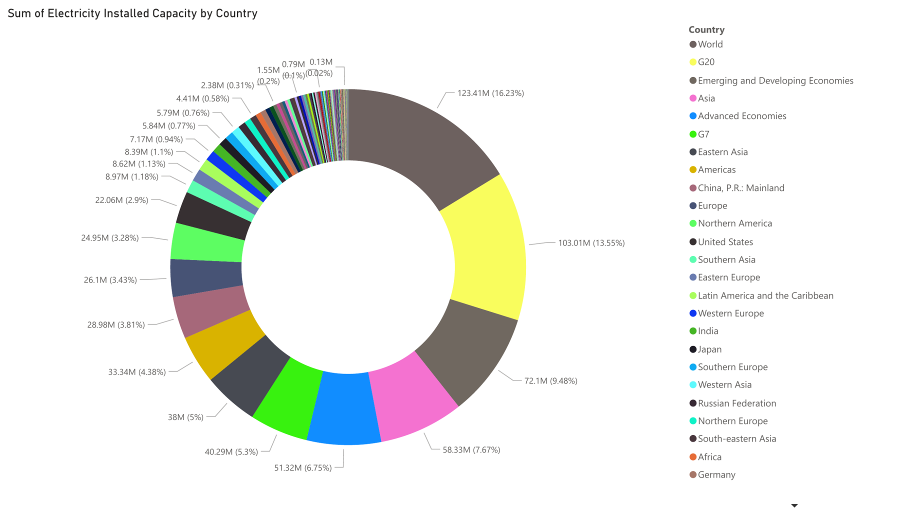


Fig. 17 Regional cumulative installed capacity share.

9.4 Prescriptive Analytics: Optimal Intervention Strategy

The overall Optimization results the effective allocation of resources prioritizes system integration over additional the capacity of expansion. The optimal investment distribution is estimated as follows: 38% for grid modernisation, 29% for battery storage, 16% for solar energy, 11% for wind energy, and 6% for hydropower.

Overall allocation of 67% towards the grid and storage infrastructure underscores the importance of integration capacity as the primary constraint in achieving efficient energy transition outcomes.

9.5 Synthesis: Performance Gap and SDG 7.2 Alignment

Our result has been consistant and indicated the overal primary barrier to achieving the main target of SDG 7.2 is not yet the scale of renewable deployment , but the overall efficiency which has been installed capacity is converted into usable electricity which is a significant improvement

Descriptive analysis describes a persistent divergence between capacity and overall output generation and diagnostic finds weak delays , regional infrastructure issue , consistent issue ,reveals weak substitution effects temporal and infrastructure delay and certain disparities from regional disputes or regime changes or even some political turmoil .predictive analytics result yield a continued growth in renewable deployment but not ready for a significant transition or switch from fossil fuels or non renewable resources . prescriptive analysis confirms that investment in overall renewable resource power plant and infrastructure and energy storage are critical for addressing system level inefficiencies

10 Discussion

The results highlight a fundamental structural issue in the global energy transition: the decoupling between renewable capacity expansion and realised electricity generation. This finding suggests that current transition strategies, which often prioritise capacity deployment, may overlook critical system-level constraints that determine actual energy output.

A key insight emerging from the descriptive and diagnostic analyses is the persistence of a capacity–generation gap. While renewable infrastructure has expanded rapidly, particularly in solar and wind technologies, this expansion has not translated proportionally into electricity generation. This discrepancy reflects the inherent intermittency of variable renewable sources, as well as limitations in grid integration and energy storage systems. Consequently, the effectiveness of renewable deployment is strongly conditioned by the ability of existing infrastructure to absorb and distribute generated power.

The diagnostic findings further indicate that the relationship between renewable capacity and fossil fuel reduction is neither direct nor immediate. The weak correlation between these variables, combined with the observed temporal lag, suggests that energy systems exhibit delayed responses to structural changes. This lag can be attributed to factors such as grid adaptation timelines, regulatory inertia, and the continued reliance on dispatchable backup generation. As a result, increases in renewable capacity may coexist with sustained fossil fuel usage over extended periods.

Regional disparities provide additional insight into the uneven nature of the energy transition. In several developing regions, capacity expansion is not matched by corresponding gains in electricity generation, indicating that infrastructure constraints play a decisive role. Limited grid connectivity, insufficient storage capacity, and operational inefficiencies contribute to this imbalance. In contrast, more developed energy systems exhibit relatively better alignment between capacity and output, although inefficiencies still persist.

The predictive analysis reinforces these observations by projecting continued growth in renewable generation alongside uneven transition trajectories. While overall electricity generation is expected to increase, the classification results suggest that not all regions are equally positioned to achieve clean energy dominance. This divergence underscores the importance of system readiness, rather than capacity alone, in determining transition success.

From a prescriptive perspective, the results clearly demonstrate that investments in grid modernisation and energy storage yield higher systemic benefits compared to additional generation capacity. The dominance of these components in the optimal allocation strategy reflects their central role in enabling efficient energy utilisation. Without sufficient grid flexibility and storage integration, additional renewable capacity risks contributing to underutilised infrastructure and diminished returns.

11 Conclusion

The analytical framework further highlights that infrastructure readiness plays a decisive role in determining transition effectiveness. Predictive results suggest continued growth in renewable generation, but with uneven progress across regions. Prescriptive analysis confirms that investments in grid modernisation and energy storage provide greater systemic impact compared to additional generation capacity, emphasizing the importance of integration over expansion.

In conclusion, the findings suggest that achieving SDG 7.2 requires a shift from a capacity-centric approach to a system-oriented strategy that prioritizes efficiency, integration, and infrastructure resilience. Addressing these factors will be essential for ensuring that renewable energy deployment translates into meaningful and sustainable energy transition outcomes.

Supplementary information. If your article has accompanying supplementary file/s please state so here.

Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

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Acknowledgements. Acknowledgements are not compulsory. Where included they should be brief. Grant or contribution numbers may be acknowledged.

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Declarations

Some journals require declarations to be submitted in a standardised format. Please check the Instructions for Authors of the journal to which you are submitting to see if you need to complete this section. If yes, your manuscript must contain the following sections under the heading 'Declarations':

- Funding : Not applicable
- Conflict of interest/Competing interests : The author declared no interest
- Ethics approval and consent to participate : Not applicable
- Consent for publication : Not applicable
- Data availability : data provided from Arena https://www.irena.org/Data
- Materials availability : Not applicable
- Code availability : analysis script are provided in the supplemenatry model
- Author contribution : All authors are contributed to study design , data analysis and manuscripts , preparations

If any of the sections are not relevant to your manuscript, please include the heading and write 'Not applicable' for that section.

below section 10 to 14 gives how to include equation, table, figure,cross refeerencing, algorithm..

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